

High Resolution Interferometric Observations to Unveil Physical Processes in Disks

Credit: NAOJ



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Cal State University Northridge

Main collaborators:

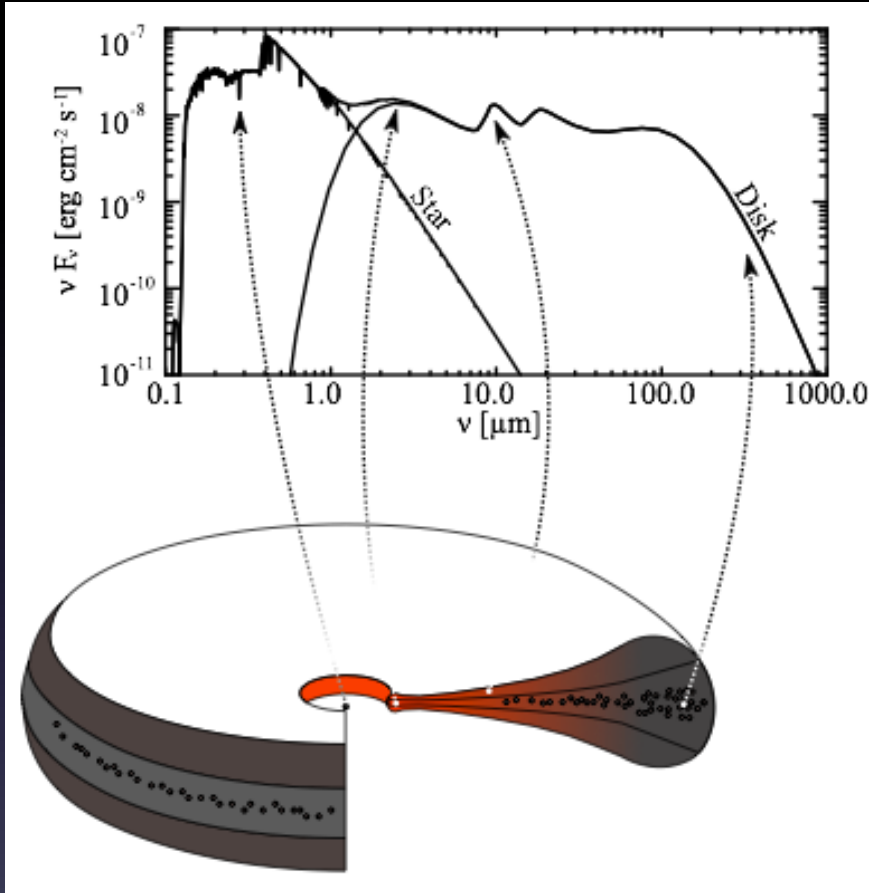
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Z. Zhu (UNLV), H. Li (LANL),
P. Barge (LAM)



Hydrodynamics of Circumstellar Disks and Planet Formation, PFITS+
June 8, 2026



Why sub-mm/mm Interferometry?



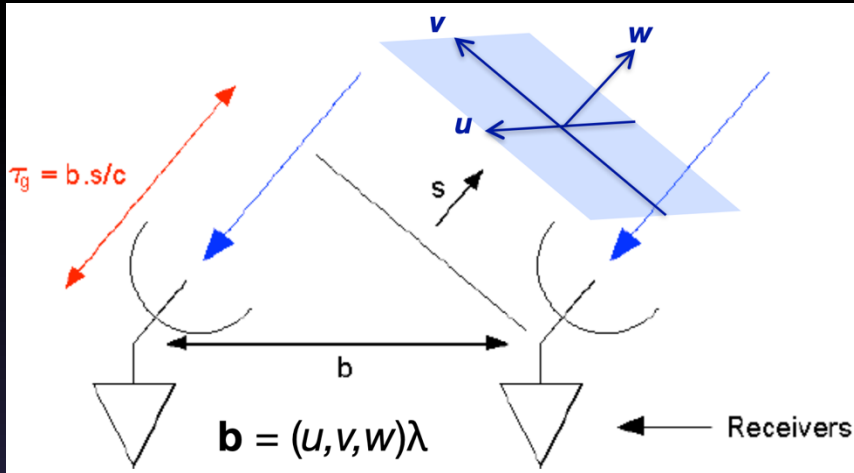
Dullemond

- Emission at low optical depths: density in disk midplane
- No contamination from central star
- Probes ~ mm-sized “pebbles”

Crash Course on Interferometry

$$I(x, y) = \iint V(u, v) e^{+2\pi i(ux+vy)} du dv$$

van Cittert – Zernike theorem

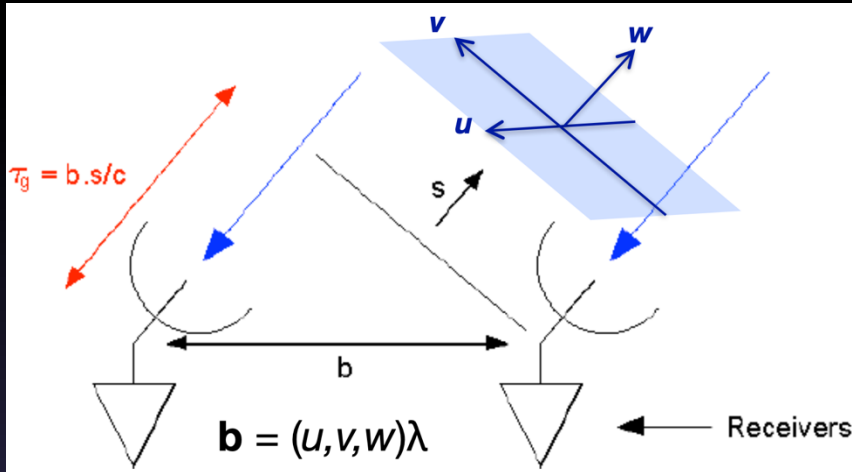


Each pair of telescopes measures $V(u, v)$ in one point of the Fourier space (u, v) = coordinates of the baseline vector $\mathbf{b}$ as seen from source

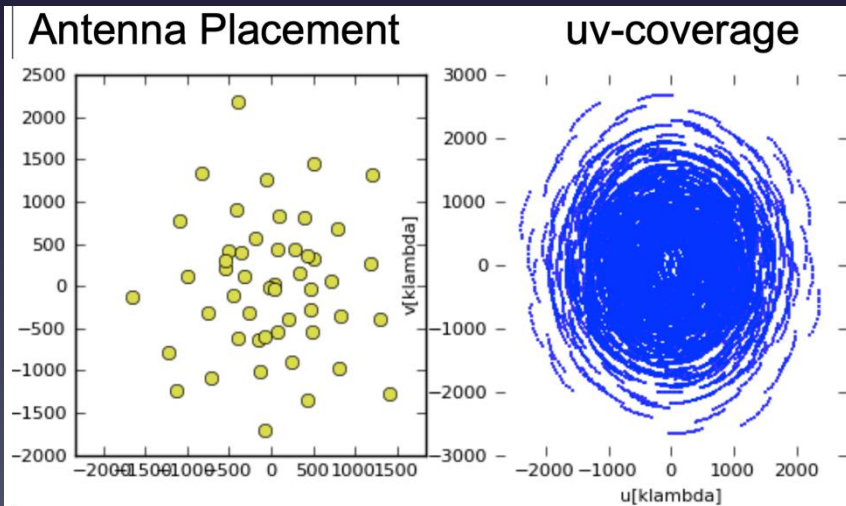
Crash Course on Interferometry

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More telescopes $\rightarrow$ better sampling of (u, v) space $\rightarrow$ better image reconstruction through $\text{FT}[V(u, v)]$; also b_{max} give information on high spatial frequencies of source emission: ang res $\sim \lambda / b_{\text{max}}$

Pre-ALMA Observations of Disks



SMA



VLA

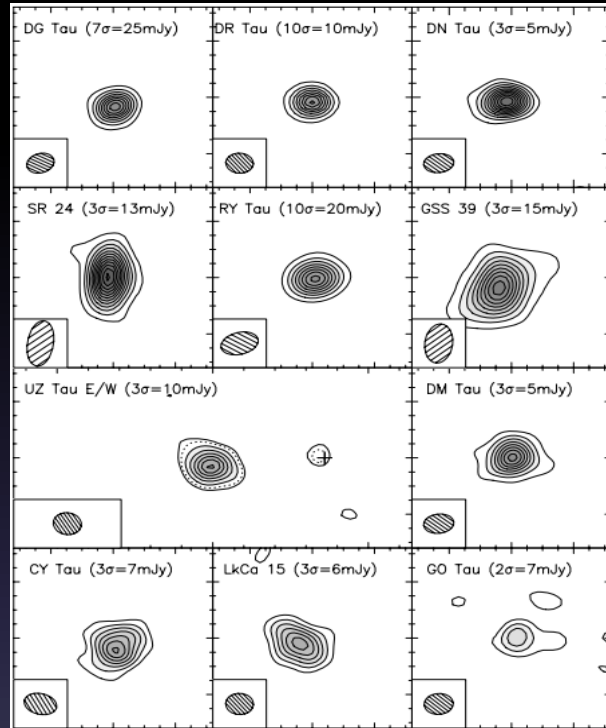


PdBI

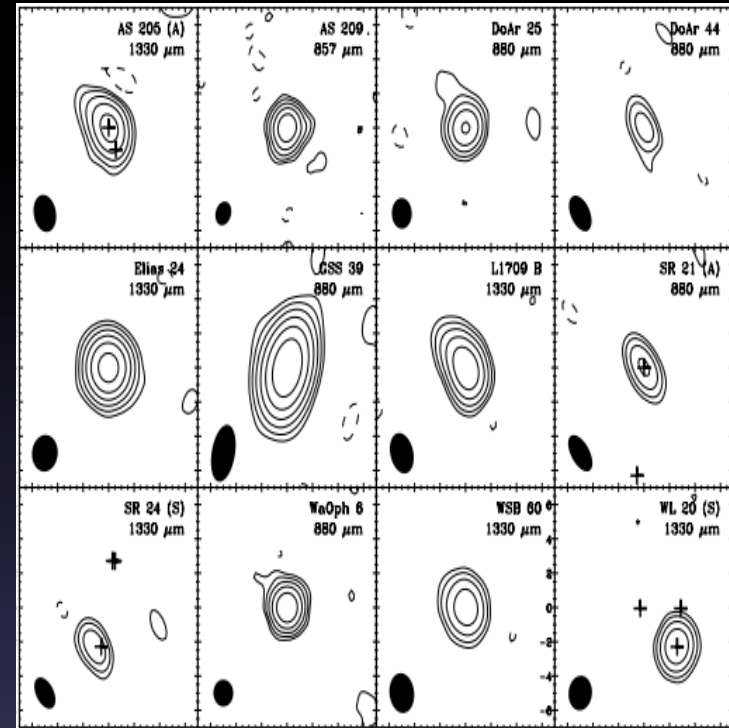


CARMA

Taurus

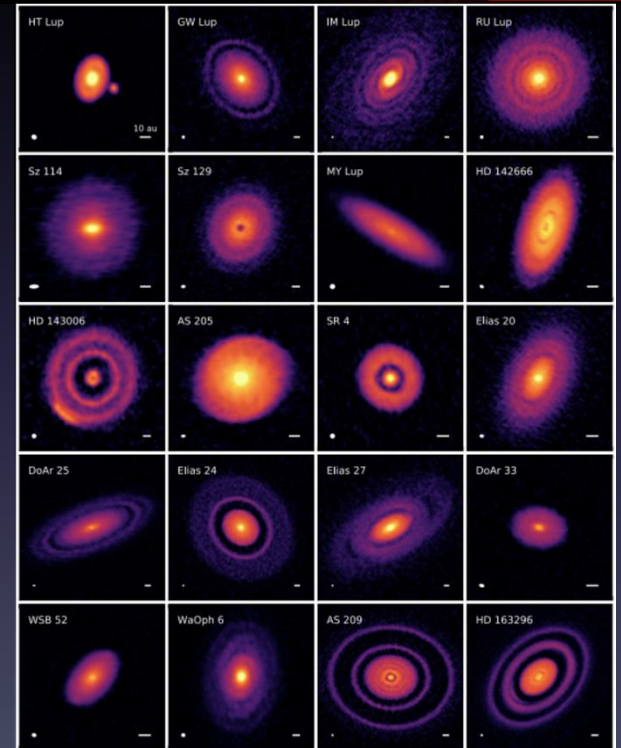
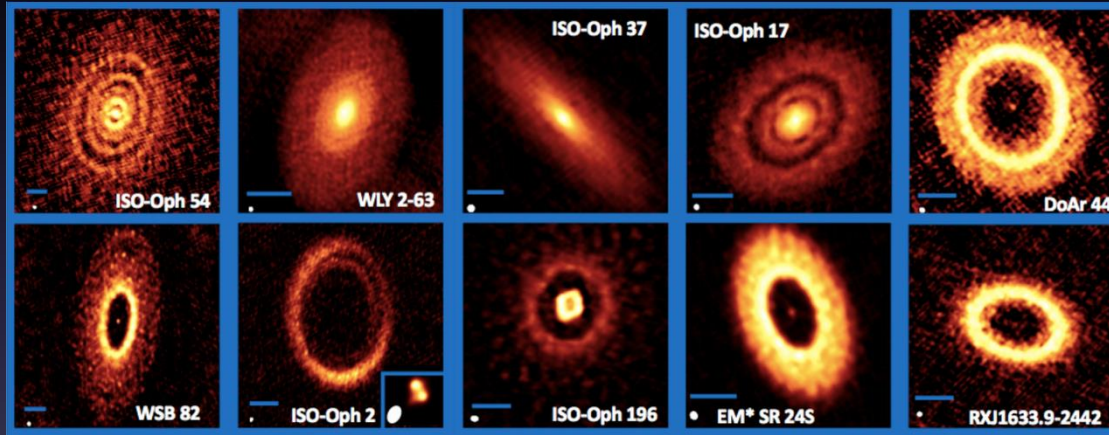
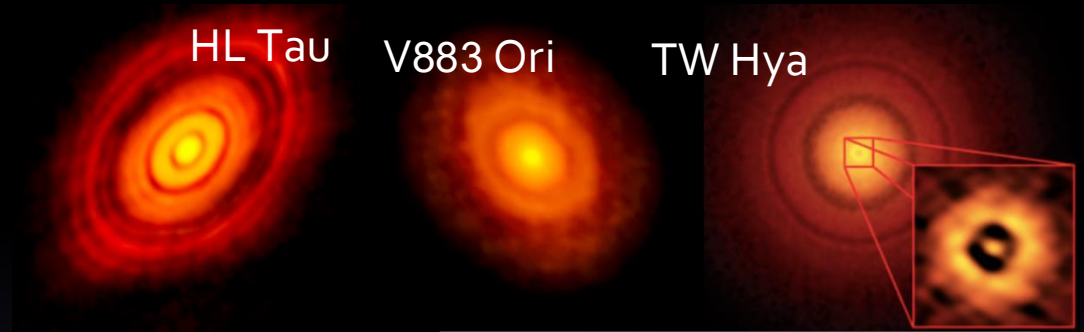
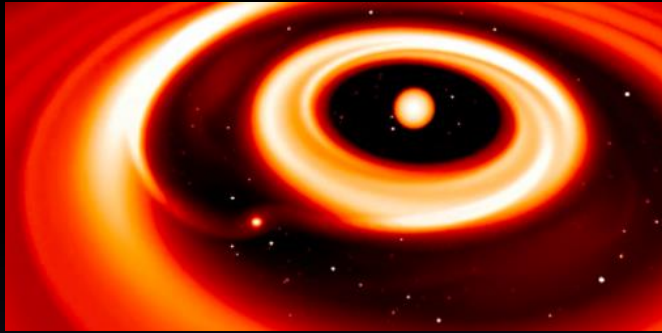


Ophiuchus



Pre-ALMA res $\sim 0.2 - 0.3''$ ($\sim 30\text{au}$):
disk substructures very rare

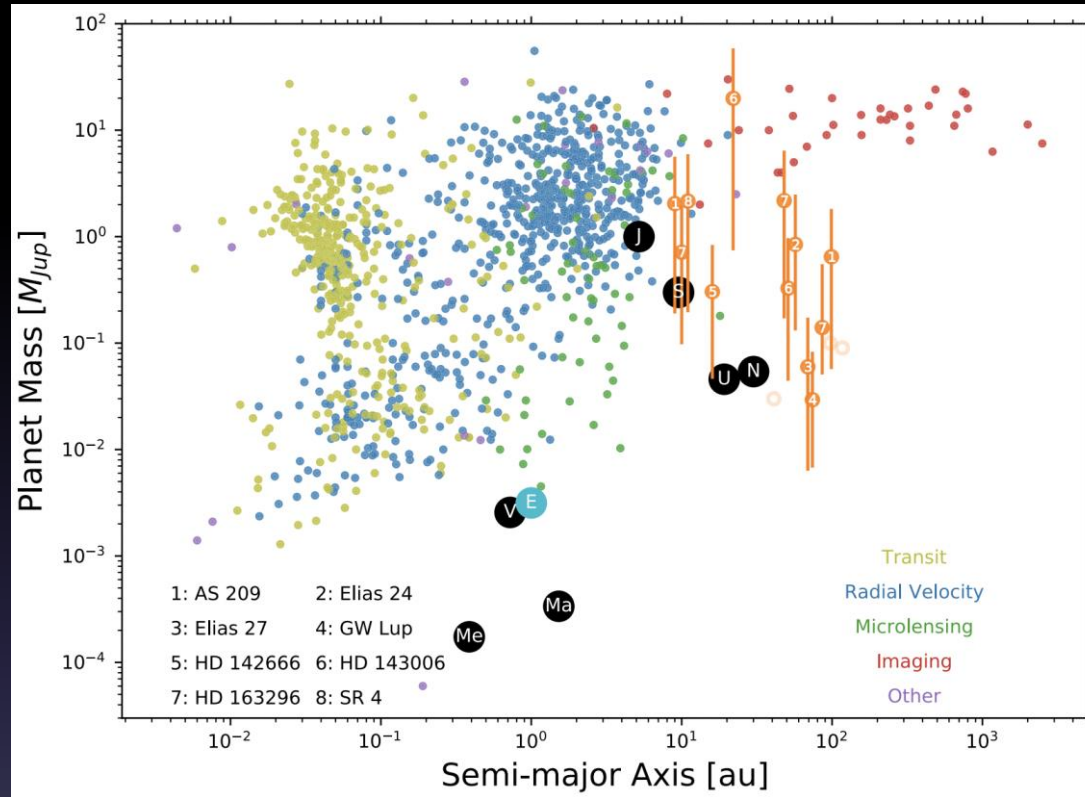
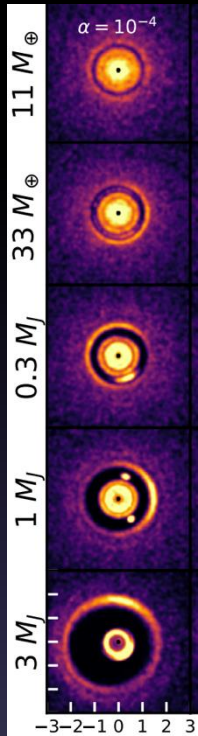
ALMA Observations of Disks



ALMA res $\sim 20 - 30$ mas (~ 3 au):
substructures in most bright disks

Young Planets from Disk-Planet Interaction

Models



Zhang, Zhu, ... Ricci+19 (see also Lodato+20)

ALMA sees disk substructures due to giant planets at $r > 10$ au

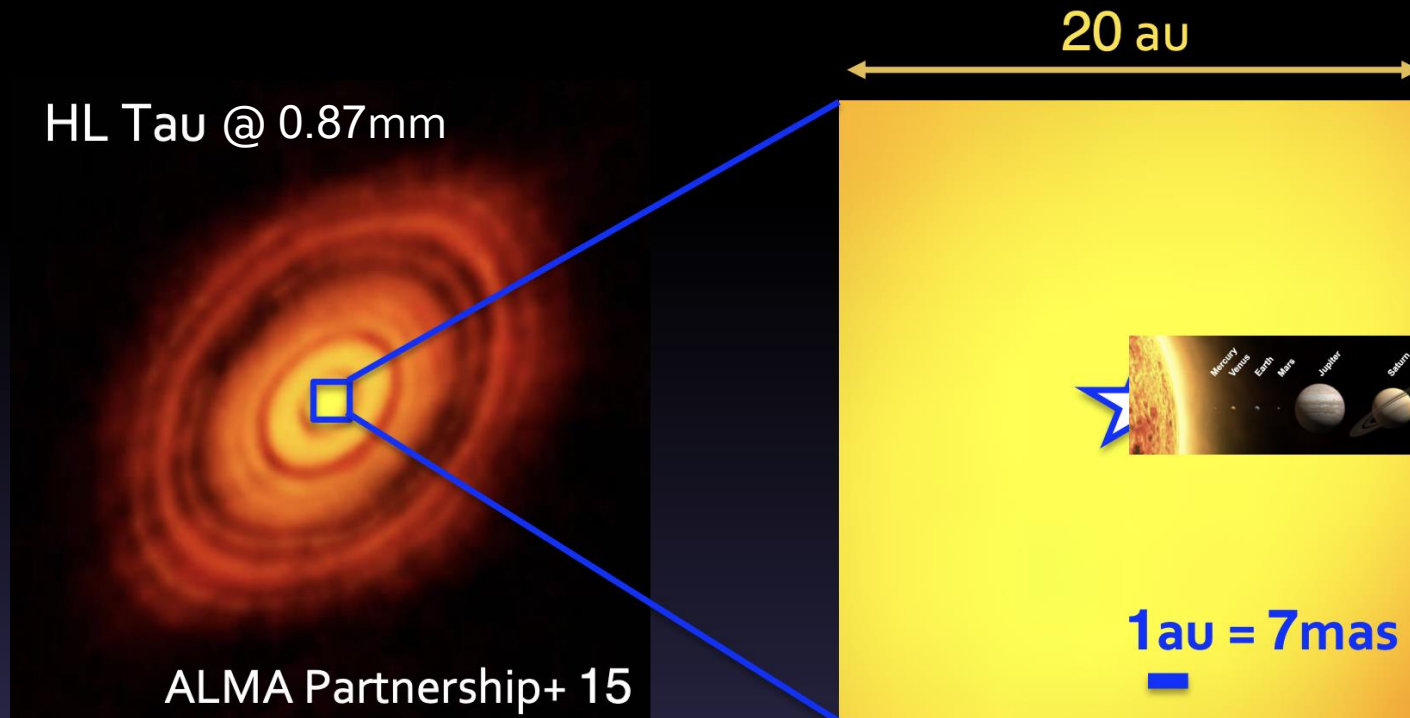
Limits of Current Disk Observations

HL Tau @ 0.87mm



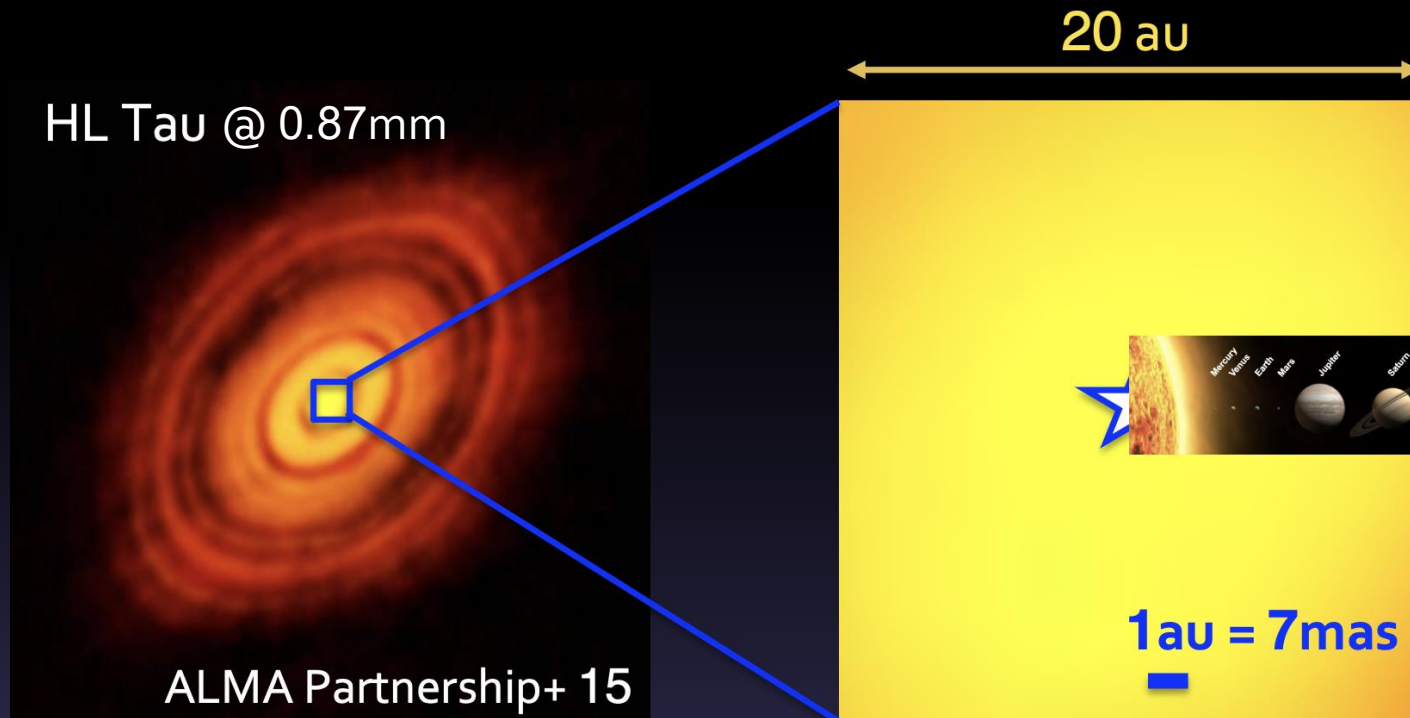
Region of terrestrial planet formation within “ALMA pixel”
Spatial resolution $\sim 5H$ at 10 au, $0.3H$ at 100 au

Limits of Current Disk Observations



Region of terrestrial planet formation within “ALMA pixel”
Spatial resolution $\sim 5H$ at 10 au, $0.3H$ at 100 au

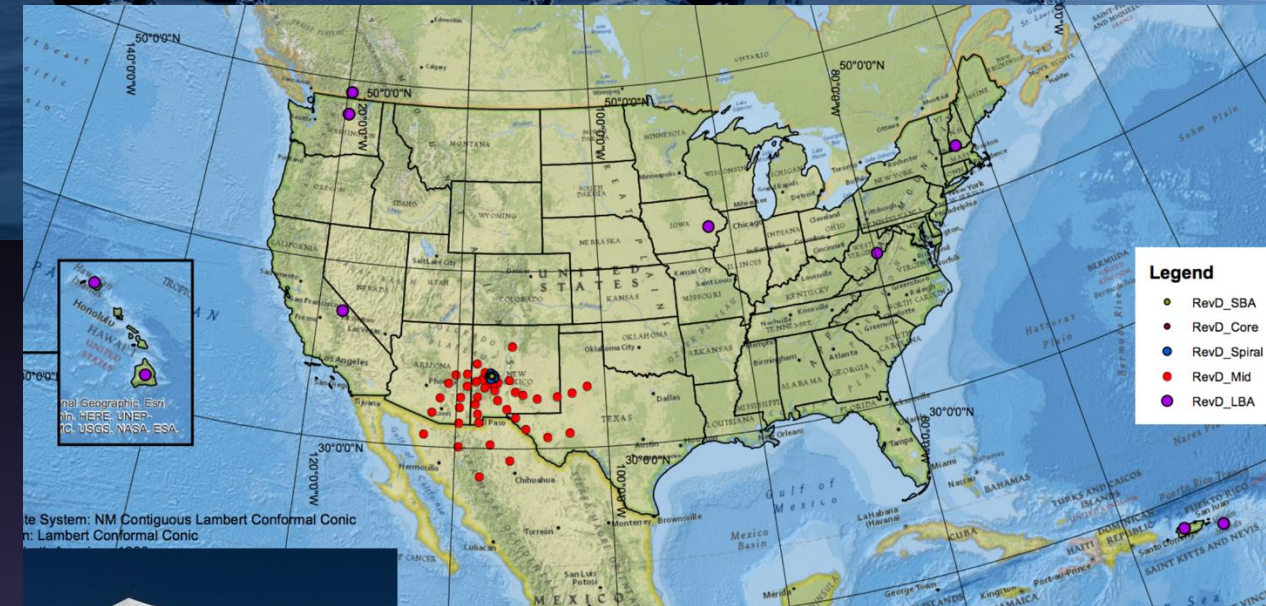
Limits of Current Disk Observations



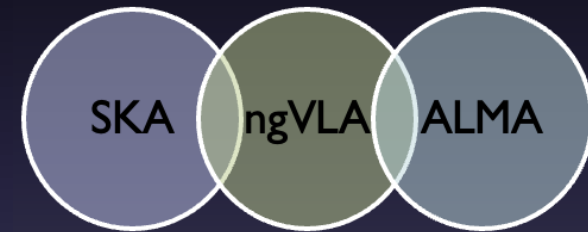
Higher ang res (< 5 mas) needed to resolve region of terrestrial planet formation & map pebbles on smaller scales
 $\lambda > 1$ mm to minimize optical depth of the dust emission

➡ Next Generation Very Large Array (ngVLA)!

Next Generation Very Large Array (ngVLA)



ngVLA bridges SKA/ALMA

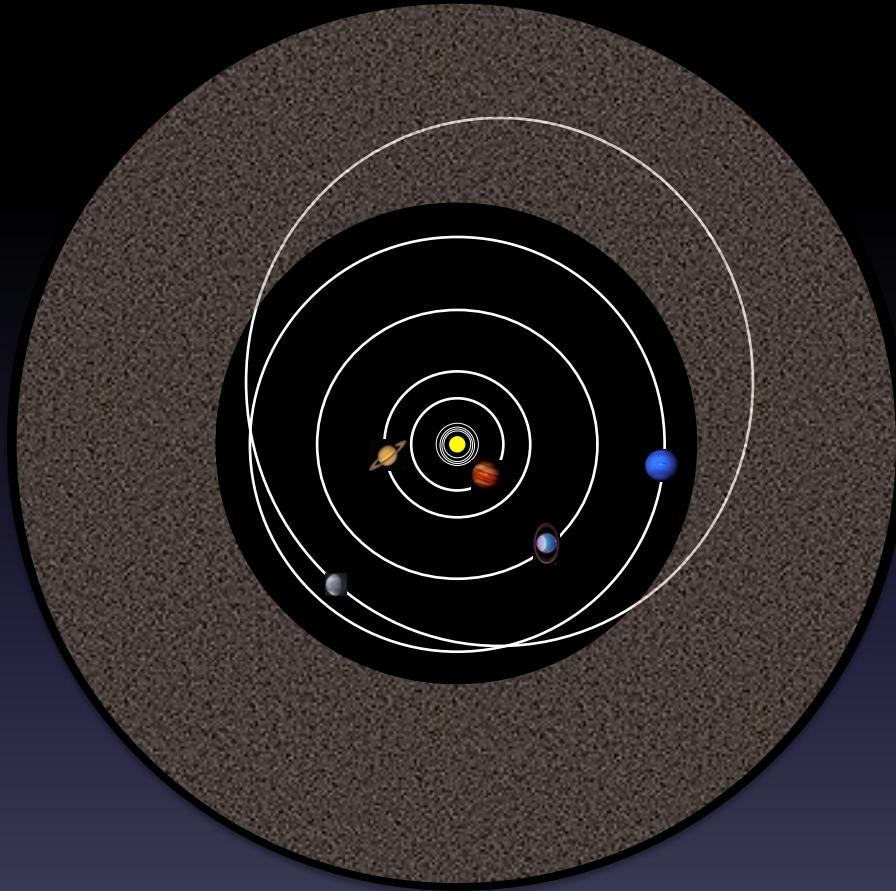


263 dish antennas at fixed locations in
US & Mexico, concentrated in Southwest;
ngVLA Early Science expected in 2032



ngVLA: the Next Revolution

At 140 pc



0.7'' resolution

0.02'' ~ 3au



ALMA

0.001'' ~ 0.1au



ngVLA

Spatial resolution ~ $0.2H$ at 10 au, $0.01H$ at 100 au

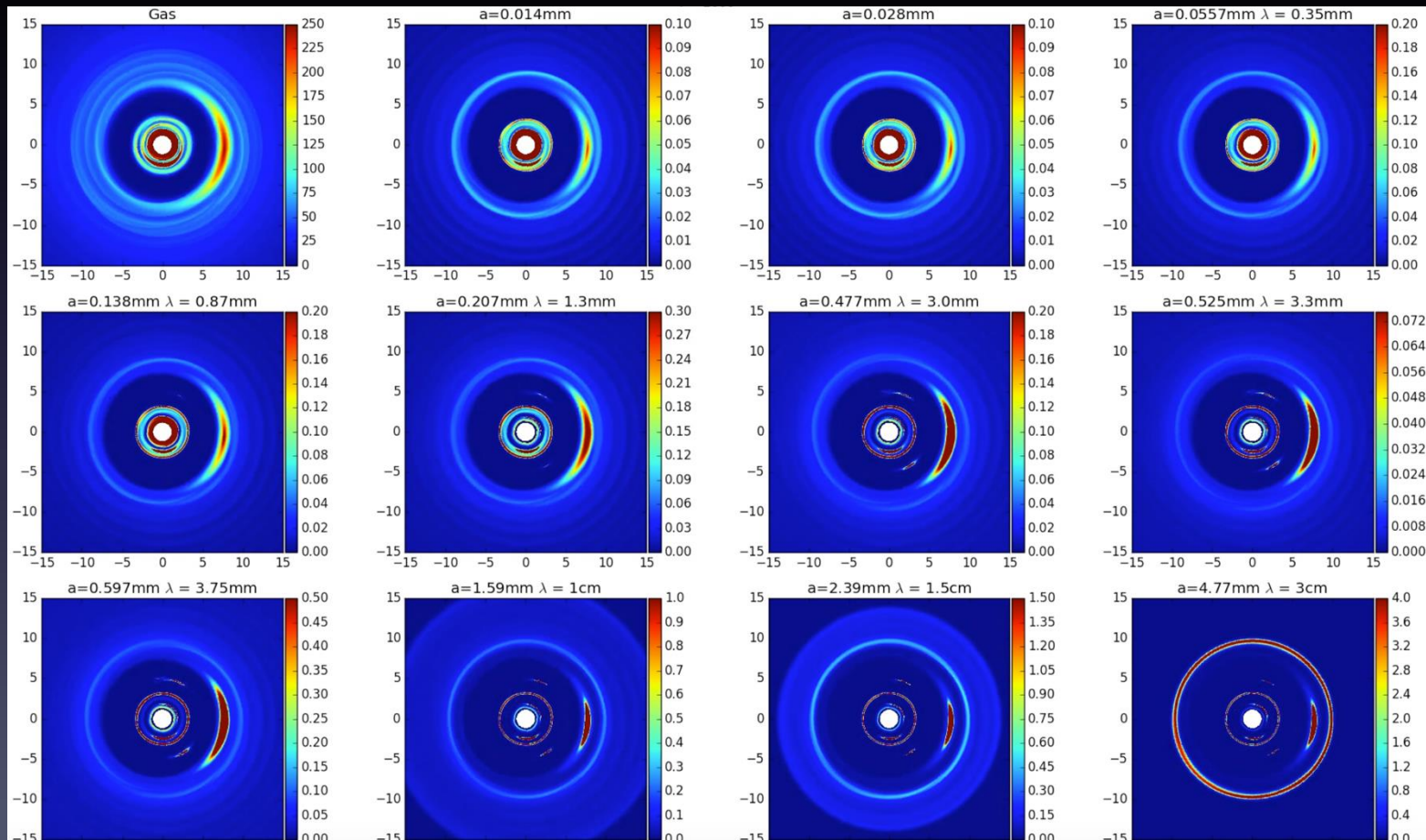
~ 10 – 100x better sensitivity than current telescopes at $\lambda > 3$ mm

Hydrodynamic Sims of Disks + Planet

LA-COMPASS bi-fluid (gas+dust) hydro code (Li+ 05, 09):

Gas evolves viscously, gravitational interaction with planet

Dust and gas coupled aerodynamically, gas drag depends on grain size



Hydrodynamic Sims of Disks + Planet

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RADMC-3D ray tracing for dust continuum (Dullemond 12)

➡ $I^{\text{model}}(x, y)$ @ 0.87, 1.3mm (ALMA), 3, 7mm, 1cm (ngVLA)

$I^{\text{model}}(x, y)$ skymodel for ngVLA & ALMA sims using
CASA simobserve & clean (see ngvla.nrao.edu)

➡ $I^{\text{obs}}(x, y)$

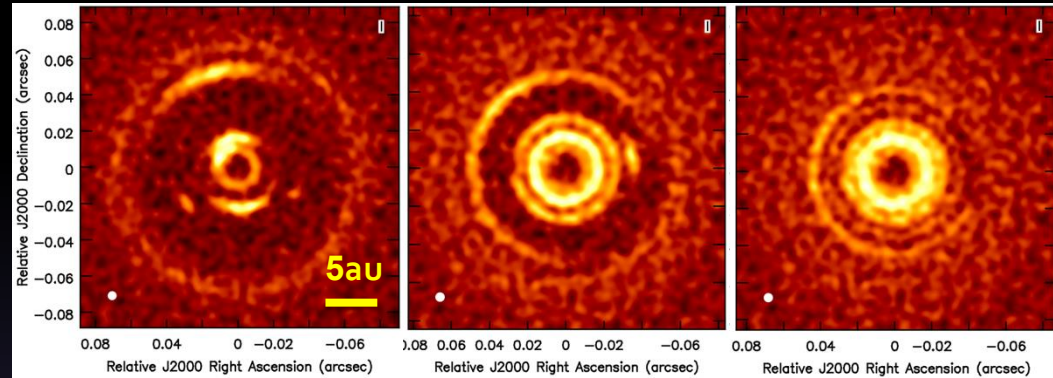
Giant Planets @ 1 - 5 au

$$M_{\text{disk}} = 0.006 M_{\text{Sun}}$$

$$d = 140 \text{ pc}$$

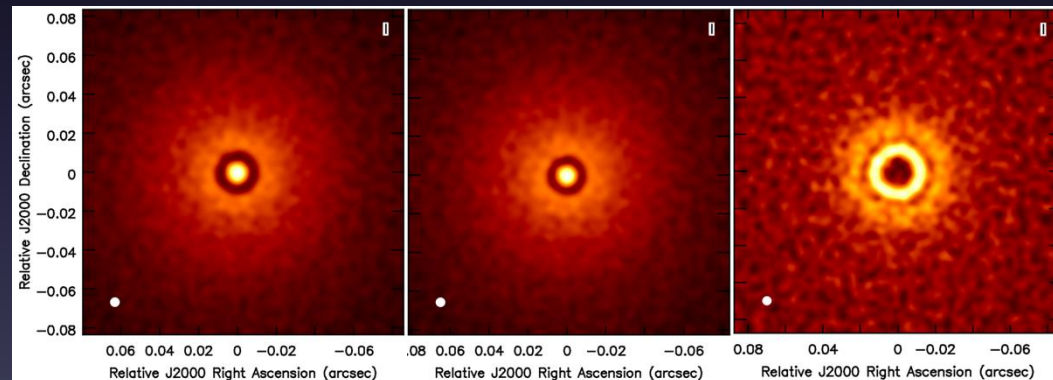
ngVLA @ 3mm
(beam = 5mas
~ 0.7 au
rms = $5e-7$ Jy/b)

Jupiter @ 5au Saturn @ 5au Neptune @ 5au



$$\alpha_{\text{visc}} = 10^{-5}$$

Jupiter @ 1au Saturn @ 1au 30 M_{Earth} @ 2.5au



$$\alpha_{\text{visc}} = 10^{-3}$$

ngVLA resolves gaps of giant planets down to ~ 1 au

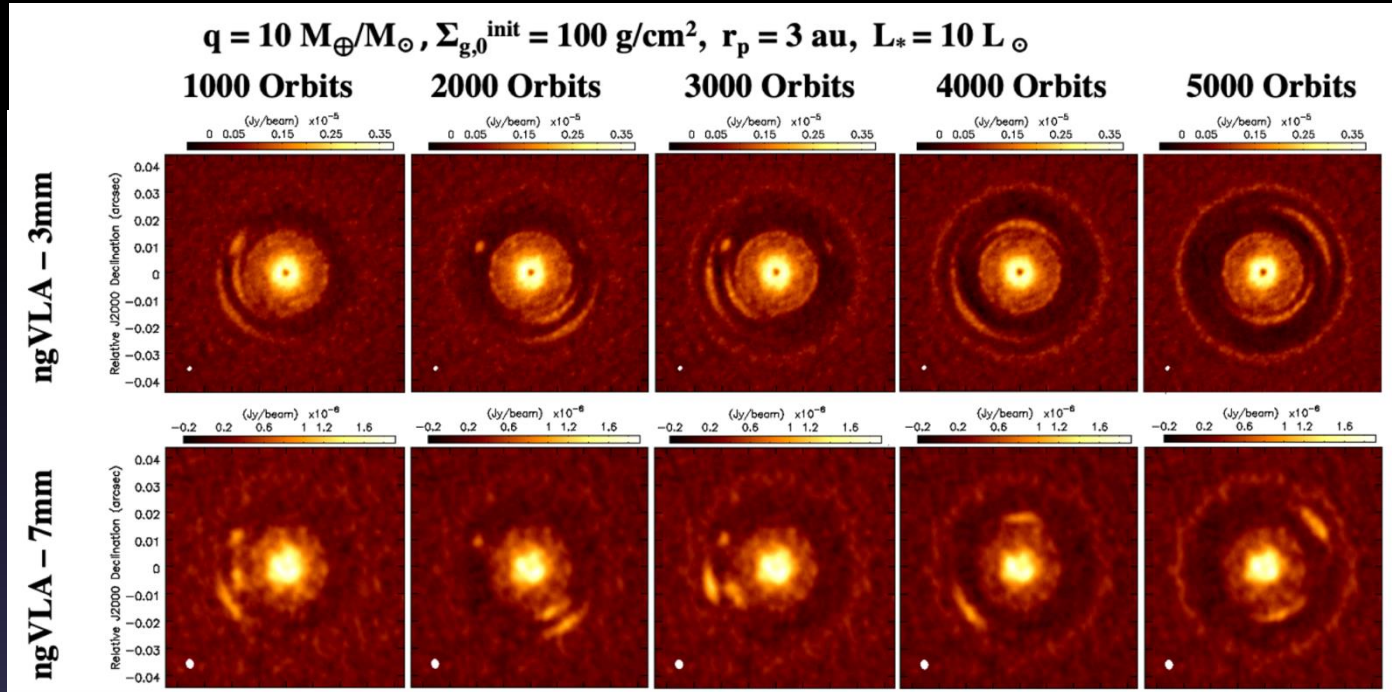
Super-Earths @ 3 au - $\alpha_{\text{visc}} = 10^{-5}$

Disk-planet sims with Dusty FARGO (Masset 2000, Baruteau & Zhu 16):

$d = 140 \text{ pc}$

ngVLA @ 3mm
(beam = 1.5 mas
~ 0.2 au
rms = $1 \times 10^{-7} \text{ Jy/b}$)

ngVLA @ 7mm
(beam = 3 mas
~ 0.4 au
rms = $5 \times 10^{-8} \text{ Jy/b}$)



ngVLA detects substructures due to Super-Earth planets around Solar-mass stars, close to Earth around low mass stars;
only way to investigate them? (Sanchis+20, Alarcon+24)

Monitoring Motions of Dust Substructures

ngVLA @ 3mm

(beam = 5mas
rms = $3\text{e-}7$ Jy/b)

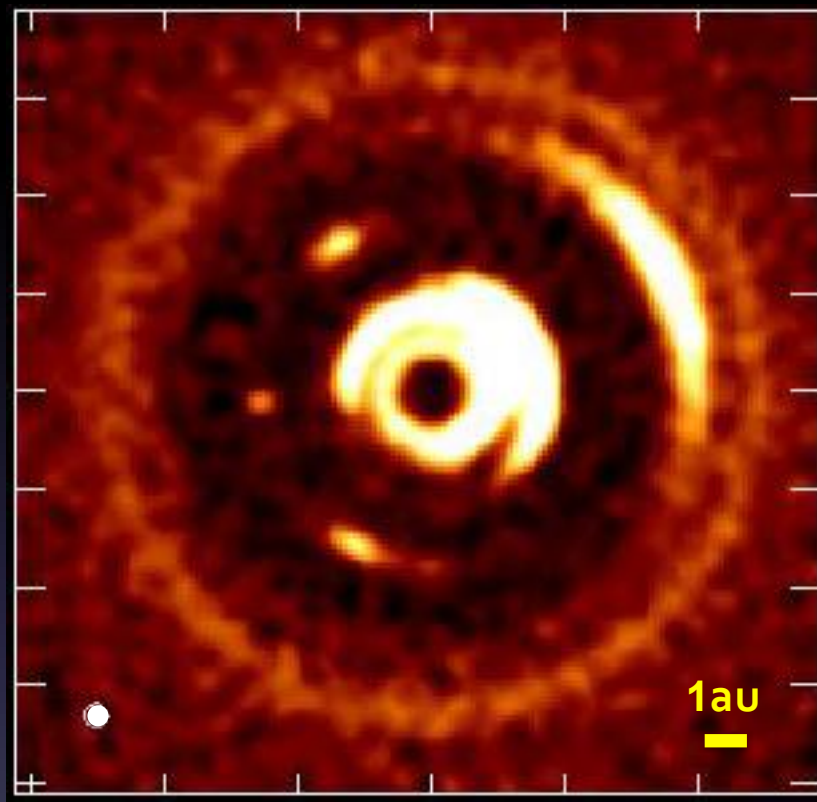
Circumplanetary disk:

$$M_{\text{disk}} = 10^{-4} M_{\text{pl}}$$

$$R_{\text{disk}} = 0.5 R_{\text{Hill}}$$

$$M_{\text{acc}} = 10^{-7} M_{\text{pl}} \text{ yr}^{-1}$$

Jupiter @ 5au, $\alpha_{\text{visc}} = 10^{-5}$



1 frame per month

1 orbit in 12 years

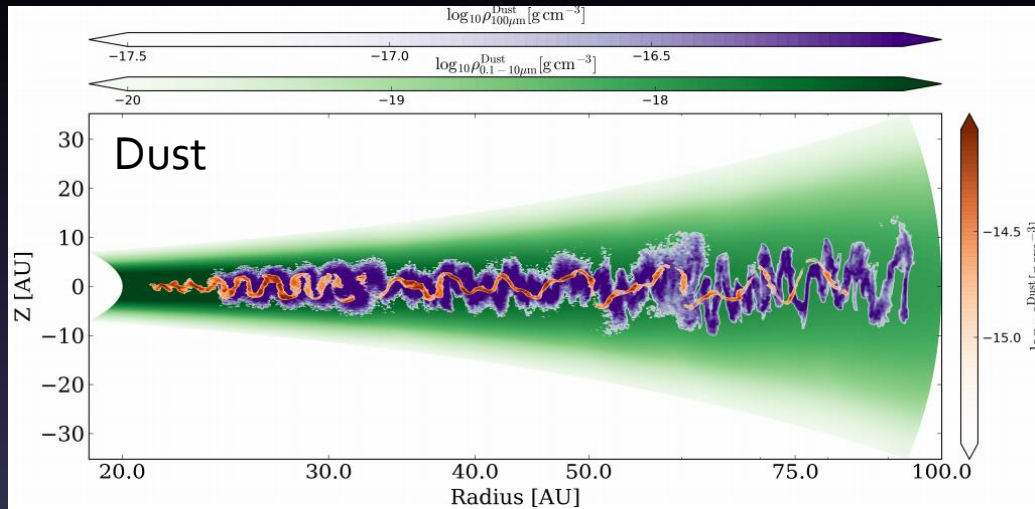
Monitor motions on timescales down to ~ weeks/months

Zhu+ 18 for ngVLA potential to detect circumplanetary disks

Ricci+ ngVLA Memo #101 for kinematic signatures in CO

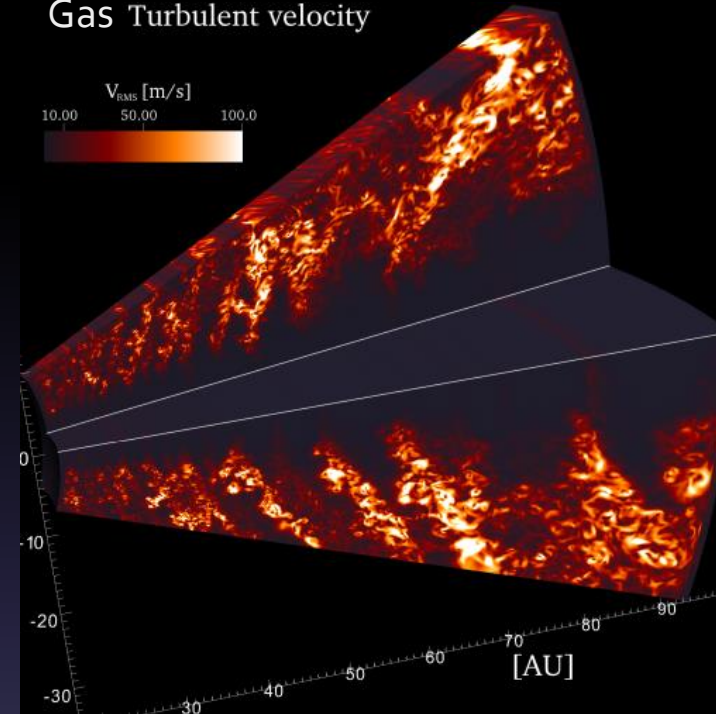
Disk Substructures due to Disk Instabilities

Ex: Vertical Shear Instabilities in
3D Radiative hydrodynamic sims



Flock+ 17

Gas Turbulent velocity



Pleiades

NASA's fastest supercomputer

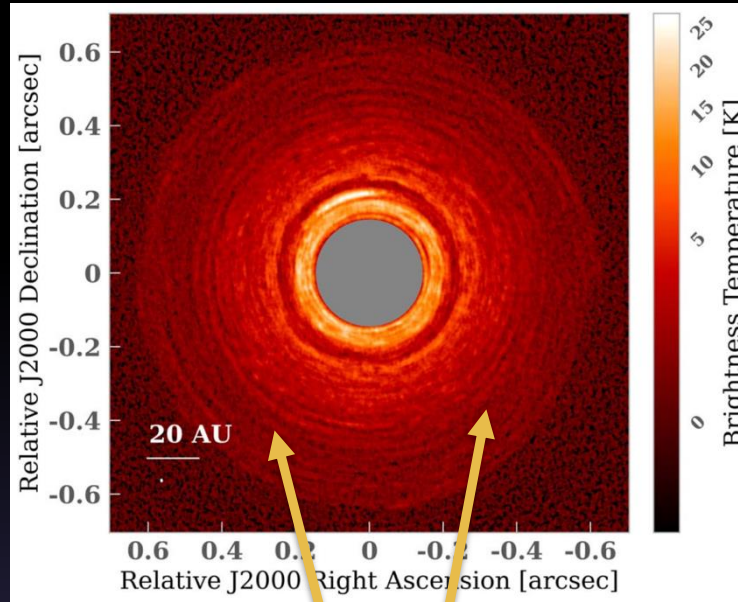


Disk Substructures due to VSI

Face-on Disk

ngVLA @ 3mm

(beam = 6mas
rms = $1\text{e-}7$ Jy/b)



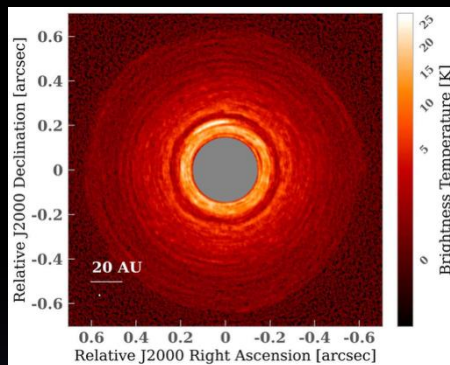
Dusty arcs of pebbles trapped
in local pressure maxima

Disk Substructures due to VSI

Face-on Disk

ngVLA @ 3mm

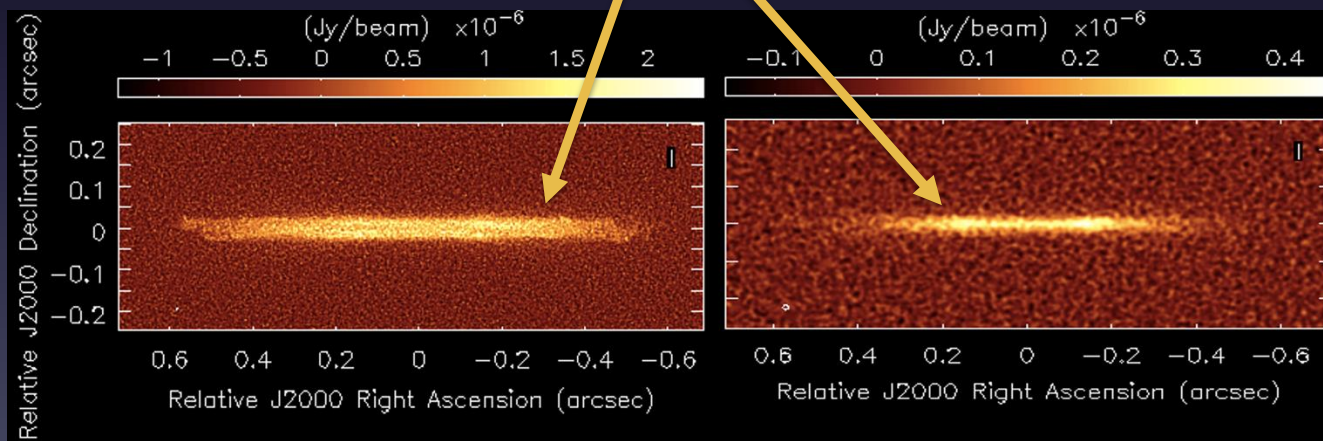
(beam = 6mas
rms = $1e-7$ Jy/b)



Vertical extent of grains with
different sizes: vertical settling

Edge-on
Disk

ngVLA
@3mm



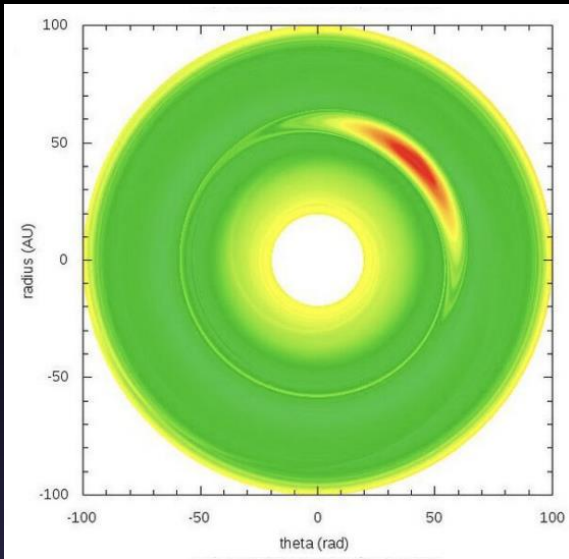
beam = 5mas
rms = $3e-7$ Jy/b

beam = 12mas
rms = $4e-8$ Jy/b

Blanco, Ricci, Flock+21

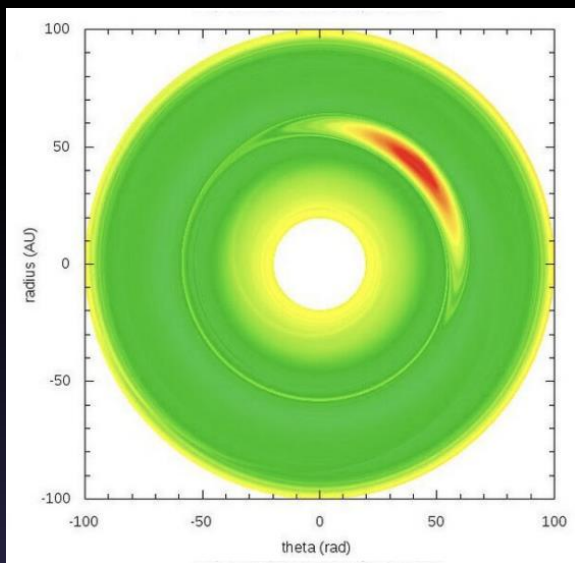
Disk Substructures due to RWI

Rossby Wave Instability produces large vortices which trap solids



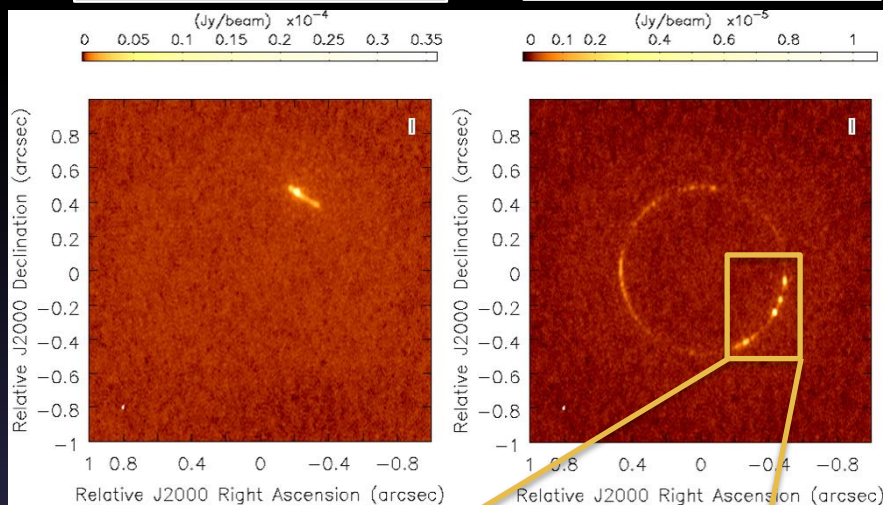
Disk Substructures due to RWI

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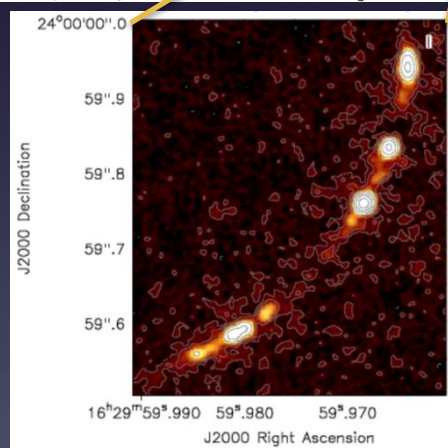


ngVLA @ 3mm

ngVLA @ 1cm



Solids concentrate in dense clumps,
potential seeds of planetesimals



Barge, Ricci+18

Carilli, Ricci+ ngVLA Memo #11

Take away messages

ngVLA key for investigating disk-planet interaction in region of terrestrial planet formation, down to Super Earths at ~ 1 au

ngVLA can resolve disk substructures due to (M)HD instabilities that affect disk physics and planet formation

ngVLA can resolve free-free emission expected for disk winds and distinguish between MHD & Photoevaporative winds (Ricci, Harter, Ercolano+ 21), as well as map dust concentration and growth across the water snowline and in the inner edge of dead zones (Ueda, Ricci, Flock+ 22)

Banzatti, Ricci+15

